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Unsupervised Machine Learning Methods

- **Clustering**
 - **What:** Groups similar data points together based on inherent patterns.
 - **Examples:** K-Means, DBSCAN, Hierarchical Clustering
 - **Applications:** Customer segmentation, social network analysis, image segmentation.
- **Dimensionality Reduction**
 - **What:** Reduces the number of features or variables while retaining key information.
 - **Examples:** PCA (Principal Component Analysis), t-SNE, UMAP
 - **Applications:** Data visualization, feature selection, noise reduction.
- **Anomaly Detection**
 - **What:** Detects rare or unusual patterns in the data.
 - **Examples:** Isolation Forest, One-Class SVM, Autoencoders
 - **Applications:** Fraud detection, cybersecurity, predictive maintenance.
- **Association Rule Learning**
 - **What:** Discovers meaningful relationships between variables in large datasets.
 - **Examples:** Apriori, FP-Growth
 - **Applications:** Product recommendations, market basket analysis.
- **Topic Modeling**
 - **What:** Extracts underlying themes or topics from large collections of documents.
 - **Examples:** Latent Dirichlet Allocation (LDA), Non-negative Matrix Factorization (NMF)
 - **Applications:** Text summarization, content categorization, sentiment analysis.
- **Autoencoders**
 - **What:** Neural networks trained to compress and reconstruct data, capturing meaningful patterns.